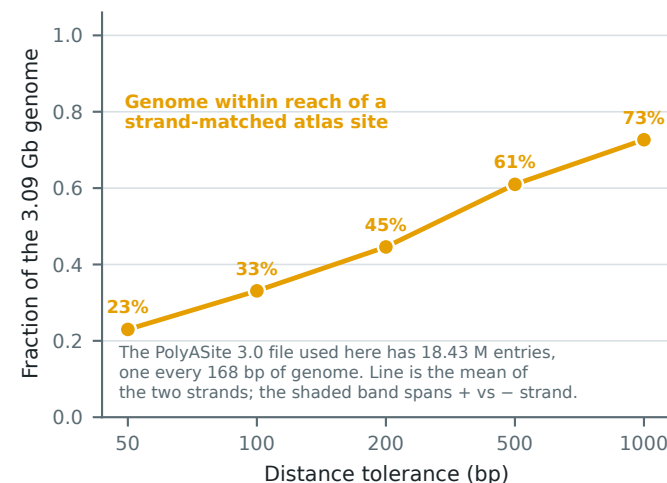
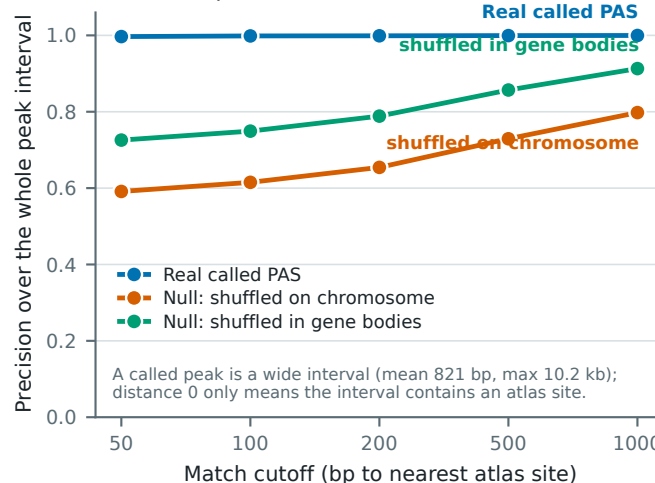


A precision of 0.9986 against PolyASite 3.0 is mostly reference density — but real PAS still beat a matched null on every test

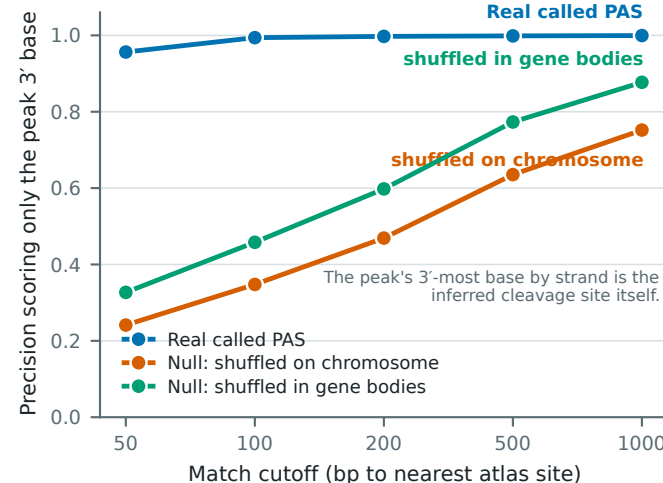
33% of the genome lies within 100 bp of a strand-matched atlas site



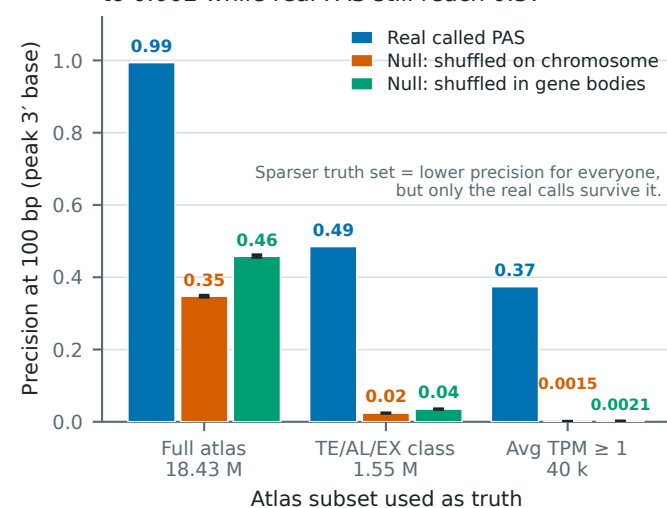
Shuffled peaks already score 0.62-0.75 when the whole peak interval is scored



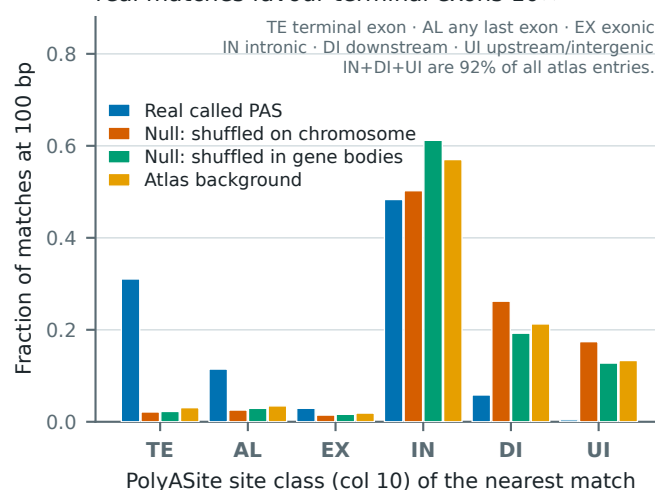
Scoring the peak 3' base alone drops the null to 0.35-0.46 while real PAS hold 0.994



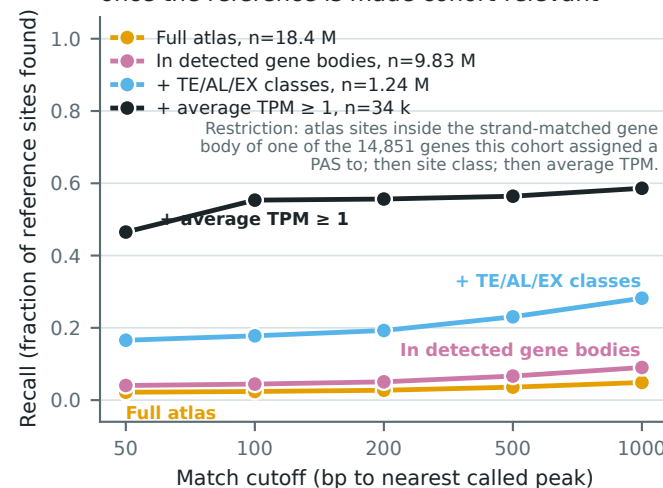
On a 40 k high-confidence atlas the null collapses to 0.002 while real PAS still reach 0.37



Null matches track the atlas background (0.7-1.1×); real matches favour terminal exons 10×



Recall at 100 bp rises from 0.024 to 0.55 once the reference is made cohort-relevant



Every panel above scores with strand-matched `bedtools closest -s -d`. The pipeline's own benchmark (`ema/benchmark/metrics.py`) does NOT require a strand match: it uses a symmetric $\pm$ cutoff window, and reports precision 0.9986 and recall 0.0326 at 100 bp on denominators 22,633 and 18,432,135. Its exact match counts are reproduced in the tsv (panel = pipeline_audit), so the strand requirement — not the 4 off-genome scaffold peaks — is why panel b reads 0.9985 and panel f reads 0.024 here. Panel f is interval-mode and so carries the same width inflation as panel b: scored on the peak 3' base its four recalls at 100 bp are 0.005 / 0.010 / 0.038 / 0.290.